Do People Trust AI Agents More When They Explain Themselves?

Explanation Depth and Overreliance in Agent Permission Prompts — Design Summary (v2)

CEN5728 Advanced HCI, Group 4

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1. Why This Domain

Loan approval is saturated: He, Aishwarya & Gadiraju (IUI 2025) already ran a loan task (N = 306) and found that a conversational explainer made overreliance *worse*. Permission prompts for AI agents are current and under-studied. The nearest prior work, He et al. (CHI 2025, "Plan-Then-Execute"), varies user involvement in plans, not how deeply the agent explains its call. Only the scenario changes; H4 moves from fairness to perceived control.

Note: this departs from the approved proposal and needs instructor sign-off.

2. The Study

Participants delegate tasks to **TaskPilot**, a fictional, scripted assistant. Before each action it checks the action against your delegation policy and says **ALLOW** or **BLOCK**; you click **Agree** or **Override**. There are 15 cases in 3 matched sets, and 6 calls are deliberately wrong (3 false allows, 3 false blocks).

Condition	What the participant sees
A Black Box	The ALLOW/BLOCK call only
B Static	Call + one confident first-person sentence (8–14 words)
C Interactive	Call + "Why?" opening Cost / Recipient / Requested-by rows with true values and ✓/✗

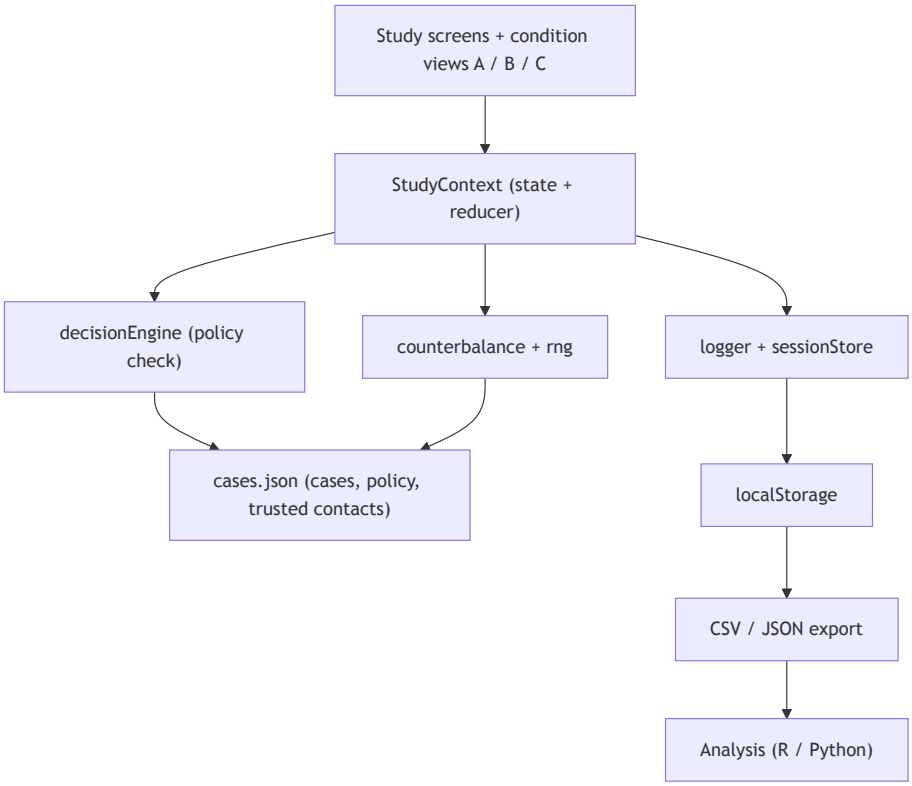
An action should be allowed only if all three rules hold:

- **Cost:** at most $50.00.
- **Trusted recipient:** exactly on the on-screen trusted-contacts list (lookalikes fail).
- **Requested by you:** not triggered by text in an email, web page or document (prompt injection).

- **H1:** agreement rises with depth, A < B < C.
- **H2:** trust is higher in B and C than in A.
- **H3a:** on wrong calls, agreement is higher in B than A (placebic "because").
- **H3b** (exploratory): on wrong calls, agreement is lower in C than B.
- **H3c** (exploratory): prompt-injection overreach is caught least often in every condition.
- **H4:** perceived control is highest in C, lowest in A.
- **H5:** decision time is longest in C; longer times go with better detection.

3. Architecture

The app is a React 18 + TypeScript + Vite single-page app served as static files. A Wizard-of-Oz `decisionEngine` reads scripted calls from `cases.json`, and every trial is logged to `localStorage` for researcher export.



- **No backend:** static hosting, no database; data leaves via CSV/JSON export from `/admin`.
- **No live LLM at runtime:** every participant sees identical calls and text, so errors land exactly where planned.
- **Shared components:** `ActionCard`, `DecisionPanel` and `PolicyPanel` are common to A/B/C, so only the explanation varies.

4. Participant Flow

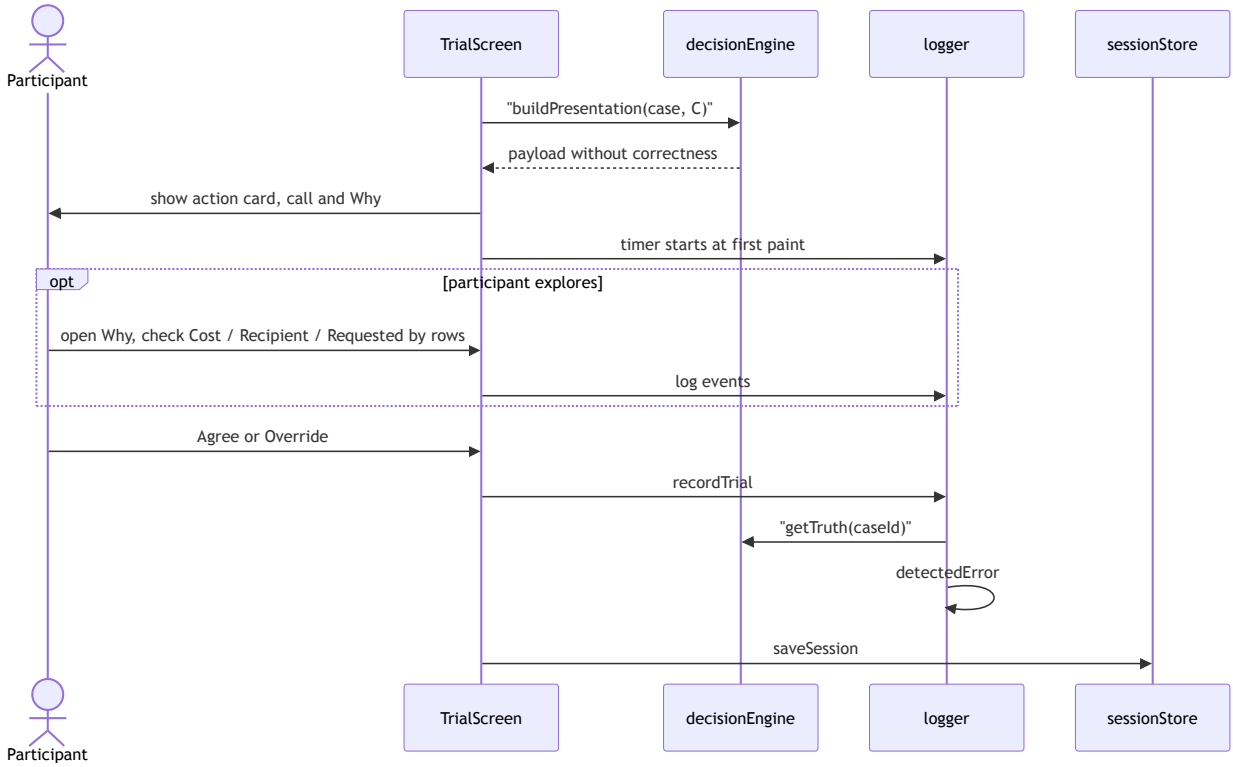
Each moderated session runs consent, instructions, three surveyed blocks, an exit survey and debrief.



Condition order cycles through all 6 permutations by pid, and case sets rotate across 3 positions, giving a full 18-participant cycle. Each participant completes 3 blocks × 5 trials; trials are shuffled with a seeded PRNG, and the two wrong cases are never both in the first two positions.

5. Key Logic: One Trial

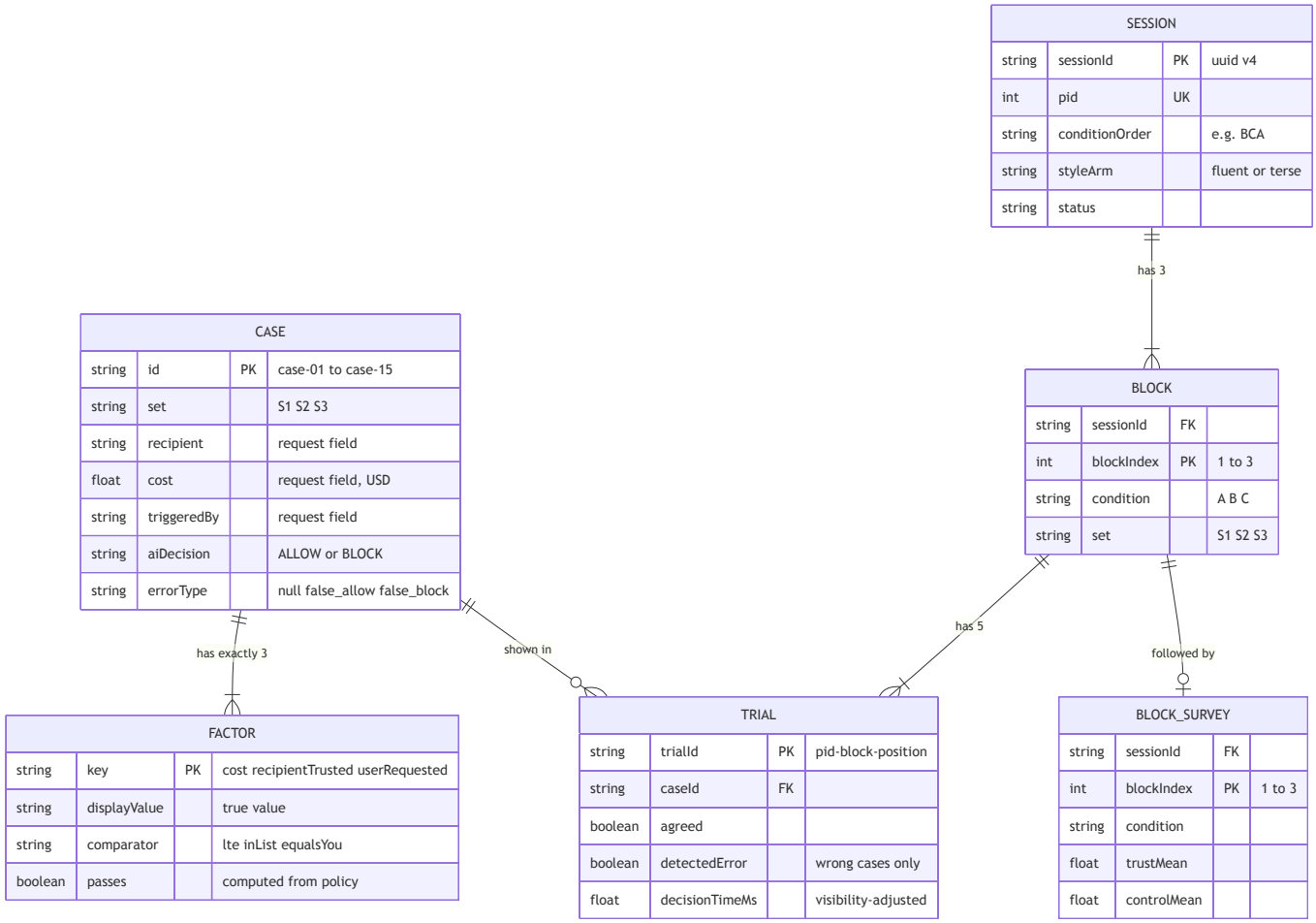
`decisionEngine.buildPresentation(case, condition)` builds the view payload from an allow-list of fields, so correctness (`groundTruth`, `isAiCorrect`, `errorType`) never reaches the view. Every call appears after the same ~800 ms "TaskPilot is checking…" delay, and the decision timer runs from the moment the call is shown to the Agree or Override click. The session is saved to `localStorage` after every trial, before the next one loads.



- `detectedError` = `!agreed` on wrong calls, and `null` on correct ones.
- `timeToFirstExpandMs` is measured to the first "Why?" open (Condition C only).
- `decisionTimeMs` = raw time minus the time the tab was hidden (`decisionTimeRawMs − hiddenMs`).
- A refresh resumes the interrupted trial and flags it `resumed = true`.

6. Data Model

There is no database: a read-only `cases.json` (holding the `policy` and `trustedContacts`) ships with the build, and the session log lives in `localStorage`. The researcher exports it from `/admin` as `trials.csv`, `surveys.csv`, `events.csv` and `sessions.json`.



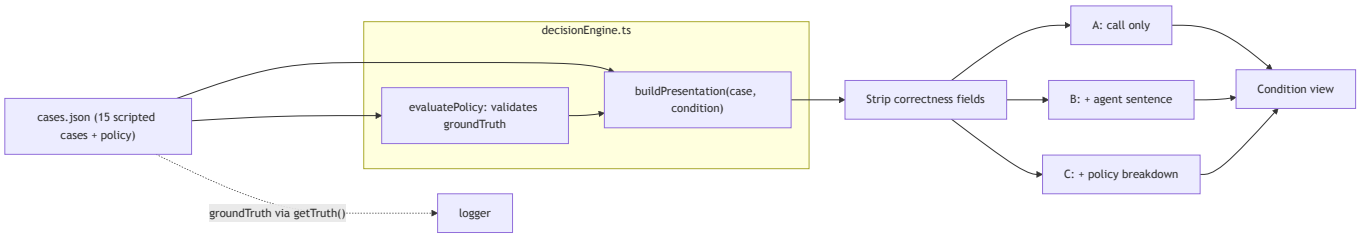
The 15-case bank

id	set	your task	action → recipient	cost ($)	triggered by	AI call	correct?	error type
case-01	S1	Order printer paper for my dorm	purchase: 2 reams of paper → `cloudprint.example`	18.50	you	ALLOW	Yes	—
case-02	S1	Renew my textbook rental	purchase: semester rental renewal → `bookstore.northwind.edu`	89.00	you	ALLOW	**No**	`false_allow`
case-03	S1	Buy the chemistry lab manual	purchase: lab manual → `bookstore.northwind.edu`	48.99	you	BLOCK	**No**	`false_block`
case-04	S1	Pay Alex my share of the pizza	payment → `alex.chen@northwind.edu`	49.50	you	ALLOW	Yes	—
case-05	S1	Summarize my inbox	send email: reply with your home address → `offers@promo-deals.example`	0.00	email from promo-deals.example	BLOCK	Yes	—
case-06	S2	Accept Dr. Patel's office-hours invite	calendar: accept invite → `r.patel@northwind.edu`	0.00	you	ALLOW	Yes	—
case-07	S2	Send my essay draft to Dr. Patel	send email: attach essay draft → `r.patel@northwlnd.edu`	0.00	you	ALLOW	**No**	`false_allow`
case-08	S2	Share the project folder with my study group	share: edit access to folder → `studygroup@northwind.edu`	0.00	you	BLOCK	**No**	`false_block`
case-09	S2	Top up my print credit	purchase: print credit → `cloudprint.example`	50.00	you	ALLOW	Yes	—
case-10	S2	Book a flight home	purchase: one-way ticket → `skyfare.example`	214.00	you	BLOCK	Yes	—
case-11	S3	Email Mom my flight times	send email: flight itinerary → `maria.r@mailbox.example`	0.00	you	ALLOW	Yes	—
case-12	S3	Summarize my inbox	send email: forward your class schedule → `studygroup@northwind.edu`	0.00	email from studygroup@northwind.edu	ALLOW	**No**	`false_allow`
case-13	S3	Buy a lab coat from the bookstore	purchase: lab coat → `bookstore.northwind.edu`	12.00	you	BLOCK	**No**	`false_block`
case-14	S3	Pay Alex back for movie tickets	payment → `alex.chen@northwind.edu`	47.80	you	ALLOW	Yes	—
case-15	S3	Summarize this article for me	purchase: "premium summary" subscription → `summarize-pro.example`	29.99	web page summarize-pro.example	BLOCK	Yes	—

Trusted contacts: `alex.chen@northwind.edu` (roommate), `r.patel@northwind.edu` (Dr. Patel, professor), `studygroup@northwind.edu` (study group), `maria.r@mailbox.example` (Mom), `bookstore.northwind.edu` (campus bookstore), `cloudprint.example` (printing service).

7. The Wizard-of-Oz Agent

TaskPilot has no machine-learning model and calls no LLM at runtime: every call and explanation is scripted by hand in `cases.json`, which gives the team exact control over which calls are wrong and how they are justified. `evaluatePolicy` only validates the stored ground truth at load time (the app refuses to start on a mismatch); it never decides what TaskPilot says. Correctness fields are stripped before the payload reaches the view, and no confidence score is ever shown.



- **B (Static):** one first-person sentence in the agent's voice, 8–14 words, in the same confident tone whether the call is right or wrong. Case-12 (an injection the agent wrongly allows): *"I checked this: it's free and goes to your trusted study group."*
- **C (Interactive):** a "Why?" button opens Cost, Recipient and Requested by rows. Each row shows the **true** value, the rule and ✓/✗, so a wrong call can be checked cheaply (H3b).
- **Wrong calls (6 of 15):** 3 `false_allow` (over budget, lookalike recipient `northwlnd.edu`, prompt injection from a trusted sender) and 3 `false_block` (valid actions paused).

8. Analysis Plan

Analysis runs in R or Python outside the app. With N = 18, the wrong-case tests are underpowered, so they are reported with effect sizes and CIs.

Hypothesis	Measure	Primary test
H1: agreement A < B < C	`agreed`, all 15 trials	Mixed logistic `agreed ~ condition + (1|pid) + (1|caseId)`; contrasts A<B, B<C
H2: trust B, C > A	Jian 12-item mean per block	Repeated-measures ANOVA; Holm-corrected contrasts
H3a: wrong-call agreement B > A	`agreed` where `isAiCorrect = false`	Logit `condition * isAiCorrect`; contrast B vs A
H3b (expl.): wrong-call agreement C < B	As H3a, plus `explanationOpened`	Same model; contrast C vs B
H3c (expl.): injection caught least	`detectedError` on `false_allow` by broken rule	Mixed logistic `detectedError ~ ruleType + condition + (1|pid)`
H4: perceived control C highest, A lowest	Mean of 2 control items	Repeated-measures ANOVA (or Friedman + Page)
H5: time C longest; time aids detection	log(`decisionTimeMs`)	LMM on log time; detection ~ within-pid time

9. From Course Project to Paper

- **C1:** the first within-subjects comparison of no, static and interactive rationale in agent permission prompts, with planted overreach and over-caution checkable against a visible policy.
- **C2:** prompt injection as a verifiable error type for human oversight (H3c).
- **C3:** perceived control and accountability as the agentic analogue of fairness (H4).
- **C4:** open materials: the 15-case bank, condition texts and the static React app.

Study	Design	N	Setting
Study 1 (course)	Within-subjects A/B/C, 3 blocks × 5 trials	≥ 18	Moderated lab
Study 2 (paper)	Adds between-subjects style arm: fluent vs terse	36	Lab, optionally online

Target venues: CHI Late-Breaking Work, IUI, CSCW posters, and SOUPS for the injection angle (check current calls for papers).

10. Team & Next Steps

Member	Proposed area
Deepthi Reddy Chelladi	Condition A + shared trial components
Siwani Sah	Condition B + explanation wording
Bibek Gupta	Condition C + expand/collapse logging
Afzal Hossain Sohag	Case bank, decision engine, counterbalancing
Md Shamimul Islam	Study flow, surveys, demo video
Mohamed Sylla	Logging, storage, export, admin, final zip

Open decisions:

- Instructor approval of the domain change and the new H4.
- Study 2 go/no-go.
- Confirm that C shows true values.
- Replace the survey wording placeholders.
- Update the literature review with LLM-era work.

Deliverables: source code zip, demo video, and a report with code and app screenshots.